

Topic 7 recap

- A **sinusoid** is a signal (voltage or current) in the form of **sine** or **cosine** function.

$$v(t) = V_m \cos(\omega t + \phi)$$

V_m : Amplitude.

$\omega = 2\pi f$: Angular frequency (*rad/s*).

$(\omega t + \phi)$: Argument of the sinusoid.

ϕ : Phase (in degrees or radians).

- A **phasor** is a **complex quantity** that represents both the **magnitude** and the **phase** of a **sinusoid**.

$$v(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m \angle \phi$$

- An electric circuit is called **AC circuit** mostly when it is operated by **sinusoidal** voltage or current **sources**.
 - They can be analyzed in what is known as **sinusoidal steady-state**.
 - All voltages and currents can be written in phasor domain rather than time domain.

Topic 7 recap

- Circuit elements have a fixed relationship between voltage and current phasors.
- Given $v(t) = V_m \cos(\omega t + \phi_v) \Leftrightarrow \mathbf{V} = V_m \angle \phi_v$ as the voltage across an element and $i(t) = I_m \cos(\omega t + \phi_i) \Leftrightarrow \mathbf{I} = I_m \angle \phi_i$ as the current through the element:

- For resistor R , voltage and current are **in phase**:

$$v = Ri \Rightarrow \mathbf{V} = R\mathbf{I} = RI_m \angle \phi_i = V_m \angle \phi_v \Rightarrow \phi_v = \phi_i$$

- For inductor L , current **lags** voltage by 90° :

$$v = L \frac{di}{dt} \Rightarrow \mathbf{V} = j\omega L \mathbf{I} = \omega L I_m \angle (\phi_i + 90^\circ) = V_m \angle \phi_v \Rightarrow \phi_v = \phi_i + 90^\circ$$

- For capacitor C , current **leads** voltage by 90° :

$$v = \frac{1}{C} \int i dt \Rightarrow \mathbf{V} = \frac{1}{j\omega C} \mathbf{I} = \frac{I_m}{\omega C} \angle (\phi_i - 90^\circ) = V_m \angle \phi_v \Rightarrow \phi_v = \phi_i - 90^\circ$$

Topic 7 recap

- **Impedance Z** of a circuit is the **ratio** of the **phasor voltage** across it to the **phasor current** through it.

$$Z = \frac{V}{I} = R(\omega) + jX(\omega) \Omega, \quad R: \text{Resistance}, X: \text{Reactance}.$$

- **Admittance Y** is the **reciprocal** of impedance.

$$Y = \frac{1}{Z} = \frac{I}{V} = G(\omega) + jB(\omega) S, \quad G: \text{Conductance}, B: \text{Susceptance}.$$

- Impedances of circuit elements:
 - For resistor R : $Z_R = R$
 - For inductor L : $Z_L = j\omega L$
 - For capacitor C : $Z_C = 1/j\omega C = -j/\omega C$
- Impedances are **combined** in series and parallel in the **same way** as **resistances** in series and parallel.
- Basic circuit laws (**Ohm's** and **Kirchhoff's**) apply to AC circuits in the same manner as DC circuits, as well as voltage and current **divisions**.
 - $V = ZI$ (Ohm's law)
 - $\sum I_k = 0$ (KCL) and $\sum V_k = 0$ (KVL)